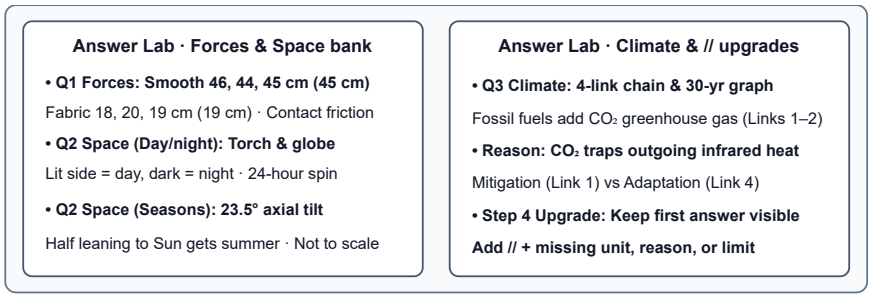
Reference sheets

Reference A · Answer Lab Bank Across Forces, Space, and Climate

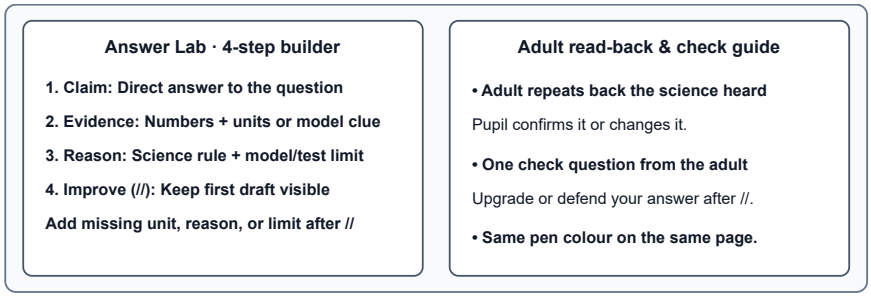


Answer Lab bank. It shows Claims, Evidence, and Reasons for Forces, Space, and Climate.

Reference only. Do not write on this sheet. Pick a question and use its Claim, Evidence, and Reason clues.

Question.	Step 1: Claim (Say it).	Step 2: Evidence (Support it).	Step 3: Science Reason & Limit.
1. Forces: Which surface lets the toy roll further?	The toy rolled further on smooth card than on rough fabric.	Smooth runs: 46, 44, 45 cm (mean 45 cm). Fabric runs: 18, 20, 19 cm (mean 19 cm).	Rough fabric has more contact friction opposing motion. Limit: 'in this test' (two surfaces).
2. Space: Why do we get day, night, and seasons?	Earth gets day and night every 24 hours, and tilted halves get seasons.	Torch-and-globe model: lit side faces torch; 23.5° tilted half leans to light.	24-hour spin gives day/night. 23.5° axial tilt during orbit gives seasons. Limit: 'not to scale'.
3. Climate: How does burning fossil fuels warm Earth?	Burning fossil fuels warms Earth's climate over 30+ years.	4-link warming chain and 30-year +1.2 °C record (not one day's weather).	Extra CO_2 greenhouse gas traps outgoing infrared heat. Mitigation (Link 1) vs adaptation (Link 4).

Reference B · Four-step Answer Lab builder and // upgrade guide



Four-step Answer Lab builder. It shows Claim, Evidence, Reason, and // upgrades.

Answer Lab step.	What to put in your answer.	Example from Question 1 (Forces).
1. Claim (Say it).	Give a clear, direct answer to the question.	'The toy rolled further on smooth card.'
2. Evidence (Support it).	Give the repeated numbers + units or the model clue.	'Smooth runs were 46, 44, and 45 cm (mean 45 cm) vs 18, 20, and 19 cm (mean 19 cm) on fabric.'
3. Reason & limit.	State the science mechanism and one model or test limit.	'Rough fabric made more contact friction pushing against the toy, in this test.'
4. Improve with //.	Keep your first answer visible. Add your upgrade after //.	'// Added cm units, the 45 cm vs 19 cm means, and contact friction.'

Made by Matt

Tier pack · Independent

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Examine the arrival drawing. Name the three parts of an Answer Lab explanation in order.

Help. Examine the headings in Box 1, Box 2, and Box 3.

2. Why is 'Smooth went 45 cm and fabric went 19 cm' stronger evidence than 'Smooth went far'?

Help. Compare a vague word with measured numbers and units.

3. Examine Box 3. Why does the scientific reason need the word friction?

Help. Think about what pushes against the rolling toy.

4. Why do we state 'in this investigation' at the end of our Forces explanation?

Help. Think about the limitation of one classroom investigation.

You do tasks

1. Use Reference A and Reference B. Construct a three-part explanation (Claim, Evidence, Reason + limit) for Question 1 (Forces) or Question 2 (Space). **Staff I ☐ S ☐ C ☐**

Help. Include the repeated measurements or model clues and the scientific mechanism.

2. Construct a second three-part explanation for Question 3 (Climate). Explain how burning fossil fuels warms Earth, and compare mitigation with adaptation. **Staff I ☐ S ☐ C ☐**

Help. Examine Row 3 on Reference A. Include Links 1 to 4.

3. Share one explanation with an adult and hear their check question. Keep your first draft and write a // improvement or defence below. **Staff I ☐ S ☐ C ☐**

Help. Examine Step 4 on Reference B.

Optional participation-loop prompt. Share one part of your science answer with an adult. Hear one check question or feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. In a Forces explanation, how do Claim, Evidence, and Reason work together?

2. In a Space or Climate explanation, what scientific mechanism belongs in the Reason step?

3. Why do we include 'in this investigation' or 'in this model' in our explanation?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Independent

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt